Project Design Phase
Solution Architecture

Date	5 February 2025
Team ID	LTVIP2026TMIDS69073
Project Name	Educational organisation using servicenow
Maximum Marks	4 Marks

Solution Architecture:

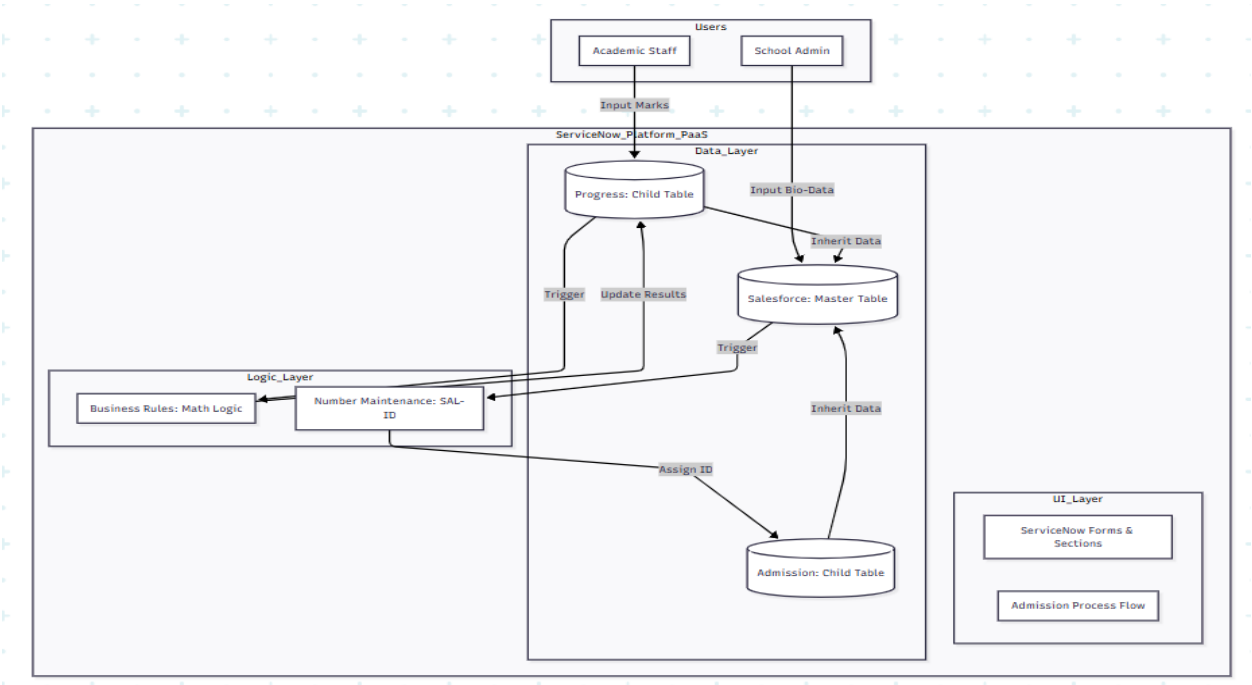
1. Architectural Overview

The solution follows a **3-tier Cloud Architecture** (SaaS/PaaS) model provided by ServiceNow. It centralizes student data, automates unique identifier generation, and performs server-side calculations to ensure data integrity across the student lifecycle.

2. Core Components of the Architecture

- **Presentation Layer (UI):** High-performance web interface using ServiceNow Form Design. It includes specialized views for Admissions (Process Flow) and Academic Progress (Grade Entry).
- **Logic Layer (Application Server):** * **Business Rules:** Server-side JavaScript that triggers the "Total" and "Percentage" calculations.
    - **Number Maintenance:** An automated engine that assigns the unique **SAL-ID** upon record insertion.
    - **Table Inheritance:** Ensures that the 'Admission' and 'Student Progress' tables extend the 'Salesforce' master table, inheriting all core bio-data automatically.
- **Data Layer (Database):** A relational MariaDB-based cloud database managed by ServiceNow, storing structured records in a parent-child hierarchy.

Solution Architecture Diagram:



4. Architecture Goals Achieved

- **Best Tech Solution:** Leveraged ServiceNow's low-code/no-code capability to build a scalable student database.
- **Defined Features:** Mapped out the "Student Registration," "Automated ID Generation," and "Grade Calculation" as distinct functional phases.
- **Structure & Behavior:** Defined the parent-child relationship between tables to ensure a "Single Source of Truth" for student data.